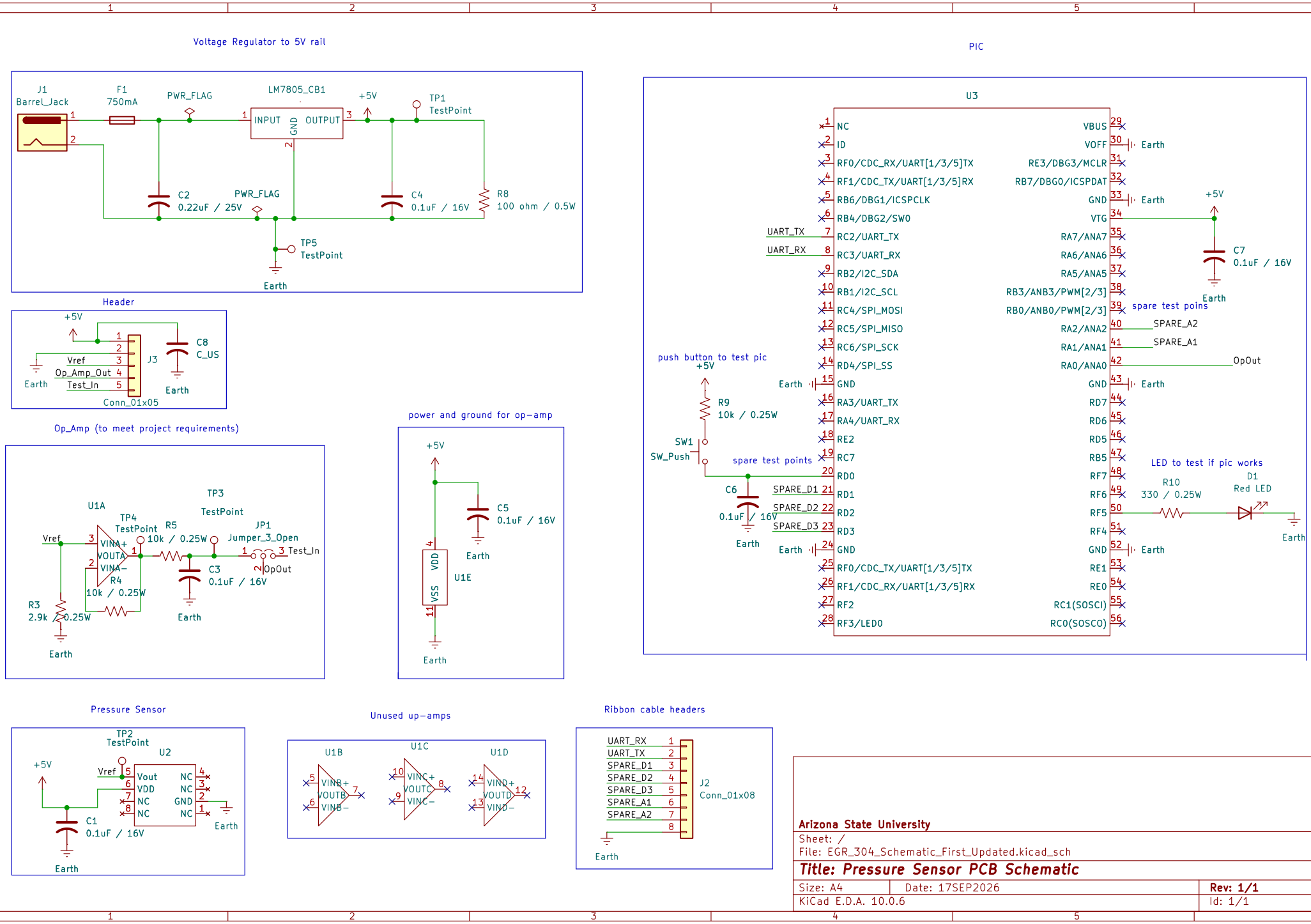

Voltage Regulator to 5V rail
J1
Barrel_Jack
F1
750mA
PWR_FLAG
LM7805_CB1
INPUT
GND
OUTPUT
+5V
TP1
TestPoint
C2
0.22uF / 25V
PWR_FLAG
C4
0.1uF / 16V
R8
100 ohm / 0.5W
TP5
TestPoint
Earth
Header
+5V
Vref
Op_Amp_Out
Test_In
J3
Conn_01x05
C8
C_US
Earth
Op_Amp (to meet project requirements)
U1A
TP3
TestPoint
TP4
TestPoint
R5
10k / 0.25W
JP1
Jumper_3_Open
Vref
VINA+
VOUTA
VINA−
R4
10k / 0.25W
Test_In
OpOut
C3
0.1uF / 16V
R3
2.9k 0.25W
Earth
power and ground for op−amp
+5V
C5
0.1uF / 16V
VDD
VSS
U1E
Earth
PIC
U3
NC
ID
RF0/CDC_RX/UART[1/3/5]TX
RF1/CDC_TX/UART[1/3/5]RX
RB6/DBG1/ICSPCLK
RB4/DBG2/SW0
UART_TX
RC2/UART_TX
UART_RX
RC3/UART_RX
RB2/I2C_SDA
RB1/I2C_SCL
RC4/SPI_MOSI
RC5/SPI_MISO
RC6/SPI_SCK
RD4/SPI_SS
GND
RA3/UART_TX
RA4/UART_RX
RE2
RC7
RD0
RD1
RD2
RD3
GND
RF0/CDC_TX/UART[1/3/5]TX
RF1/CDC_RX/UART[1/3/5]RX
RF2
RF3/LED0
VBUS
VOFF
Earth
RE3/DBG3/MCLR
RB7/DBG0/ICSPDAT
GND
Earth
VTG
+5V
C7
0.1uF / 16V
RA7/ANA7
RA6/ANA6
RA5/ANA5
RB3/ANB3/PWM[2/3]
RB0/ANB0/PWM[2/3]
spare test poins
SPARE_A2
SPARE_A1
RA2/ANA2
RA1/ANA1
RA0/ANA0
OpOut
GND
Earth
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
GND
Earth
RE1
RE0
RC1(SOSCI)
RC0(SOSCO)
push button to test pic
+5V
R9
10k / 0.25W
SW1
SW_Push
spare test points
C6
0.1uF / 16V
Earth
SPARE_D1
SPARE_D2
SPARE_D3
LED to test if pic works
R10
330 / 0.25W
D1
Red LED
Earth
Pressure Sensor
TP2
TestPoint
U2
+5V
Vref
Vout
VDD
NC
NC
NC
NC
GND
NC
C1
0.1uF / 16V
Earth
Unused up−amps
U1B
U1C
U1D
VINB+
VOUTB
VINB−
VINC+
VOUTC
VINC−
VIND+
VOUTD
VIND−
Ribbon cable headers
UART_RX
UART_TX
SPARE_D1
SPARE_D2
SPARE_D3
SPARE_A1
SPARE_A2
J2
Conn_01x08
Earth
Arizona State University
Sheet: /
File: EGR_304_Schematic_First_Updated.kicad_sch
Title: Pressure Sensor PCB Schematic
Size: A4
Date: 17SEP2026
Rev: 1/1
KiCad E.D.A. 10.0.6
Id: 1/1